A Major Project Report
(Mid Term Evaluation)

On

Sentinel-2 multispectral change detection for monitoring urban expansion and loss of water bodies

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1. Current Implementation / Prototype Status

Within the first two weeks of project implementation (Week 1 and Week 2), the team successfully developed the core computational infrastructure, data collection mechanism, and prototype of the project. The aim of the project was to study the process of urbanization and the depletion of the surface water resources in Greater Noida (Uttar Pradesh, NCR) and Bengaluru Urban (Karnataka) over the 8-year period (2018 – 2026).

The automated processes of the Google Earth Engine (GEE) are completed and are available for integration into the modular infrastructure, which is saved in the project database. In particular, the pipeline will be able to communicate with the giant COPERNICUS/S2_SR_HARMONIZED Sentinel-2 Level-2A surface reflectance virtual data storage. The procedure of the cloud/shadow detection will include the use of the QA60 bitmask (clouds with the highest density) and Scene Classification Layer (SCL) to filter the atmospheric turbulences.

For the execution of the calculations of the indices, the server unit uses the matrix computing for all the important indices for the pre-monsoon (March-May) and post-monsoon (October-December) periods from 2018 till 2026.

- Normalized Difference Vegetation Index (NDVI = (B8 - B4) / (B8 + B4)) – vegetation index that measures vegetation health and distinguishes vegetation coverage from bare soil.
- Modified Normalized Difference Water Index (MNDWI = (B3 - B11) / (B3 + B11)) – used for identification of visible standing water on the ground surface, minimizing the effect of built-up environment.
- Normalized Difference Built-Up Index (NDBI = (B11 - B8) / (B11 + B8)) – applied, it identifies areas of impervious built-up.
- The Streamlit will be set up in order to build a dashboard with the following possibilities of the users' interactive analysis of the trajectories of the area in real time: spatial matrix analysis, creation of change matrices, validation.

A prototype Streamlit decision-support dashboard is being established with four distinct analytical views, allowing real-time inspection of area trajectories, spatial matrix comparisons, change matrices, and validation metrics.

2. Design / Architecture & Validation

The software architecture follows a rigorous two-tier geospatial design pattern, separating petabyte-scale cloud raster processing from analytical tabular reporting:

• Tier 1 (Cloud Raster Engine - Google Earth Engine): Handles multispectral image collection, cloud masking, band scaling, seasonal compositing, and pixel-level Random Forest classification across district boundaries.

• Tier 2 (Analytical Database & Dashboard): Exports flat-file database tables (area_stats.csv, area_stats_raw.csv, accuracy.csv, change_matrix.csv) and populates an in-memory SQLite relational database (pt377_geospatial.db) containing pre-built SQL views (vw_urban_water_comparison).

Methodological Diagnosis and Patched v2 Architecture

During the initial mid-term validation, one critical technical issue has been identified by the team: processes of training of the main single-year classifier exhibited abnormal land cover behavior (such as, for example, reduction of the built-up areas from 7% to 26% and 62.5% instabilities in Greater Noida). Through a thorough diagnostic analysis, it has been identified that the cause of such behavior is annual variability in the phenology and different decision threshold between dry bare soil and built-up area.

In order to deal with the situation, a new structure has been developed and tested by the company that is currently used in the evaluation phase:

- Individual Anchor Random Forest Classifier 2021: A single random forest classifier (consisting of 100 different trees) is developed based on the corresponding samples collected in all 5 cities and in all the seasons during the target year 2021, comparing with ESA's WorldCover v200 (where Class 50 is Built-up and Class 80 is Water); and the same approach is repeated for all the other years from 2018 to 2026.
- 3-Year Periodic Mainstream Filter: Uses a 3-year sliding nominal seasonal mode filter ([t-1, t, t+1]) for eliminating inaccuracies due to the misclassification occurring only in a single year.
- Built-up No Reversion Constraint: Ensures that an area which has been urbanized (Class 0) will not revert back to its original natural cover within 8 years.

3. Testing / Experimental Results So Far

The tests have been carried out for the regions of Greater Noida (Gautam Buddha Nagar District, UP) and Bengaluru Urban (Karnataka) from 2018 to 2026. There are plenty of automated unit tests (9 passing tests in test_pipeline_and_dashboard.py file) as well as sanity check Step-2 tests proving that the schema is fully complete.

Mid-Term Empirical Findings (Post-Monsoon Season)

• Greater Noida: Built-up Area increased from 118.80 km2 in 2018 to 174.40 km2 in 2026 (+46.80%). Water area decreased from 19.40 km2 to 15.56 km2 (-19.79%), showing an inverse relationship (r = -0.998).

• Bengaluru Urban: Built-up Area increased from 383.00 km2 in 2018 to 468.60 km2 in 2026 (+22.35%). Stream water area decreased from 31.90 km2 to 25.34 km2 (-20.56%).

• Accuracy Assessment: 2021 hold-out classification achieved an Overall Accuracy of 85.19% (Kappa = 0.8020), whereas 2020 transferability test resulted in 81.92%.

4. Supporting Evidence / Repository / Demonstration Links

• GitHub Repository: https://github.com/devanshiSaini06/26P_3CSE377

• Local Execution Scripts: Modular execution via run_pipeline.sh / run_pipeline.bat, with automated test discovery via python -m unit test discover -s tests -v.

5. Major Challenges / Pending Work (Optional)

• Challenge Addressed: Removing confusion between dry land, soil and built up High roofs in North India in Monsoon and Pre Monsoon season by using strict ESA World Cover by using the constraint variable B2>0.085 as the algorithm considered high roofs as high lands.

• Pending Work for Phase III: Collecting independent validation points by using over 160 data points in Google Earth Engine Dataset of Sentinel-2. Making dashboard, rectifying covariance for project report compilation and completion.